

Inductive Principle over $\Downarrow_e$

$$\forall B_1, B_2 \in \text{Bool}. [Q(B_1, \text{false}) \implies Q(B_1 \& B_2, \text{false})]$$

$\wedge$

$$\forall B_1, B_2 \in \text{Bool}. \forall b \in \mathbb{B}. [Q(B_1, \text{true}) \wedge Q(B_2, b) \implies Q(B_1 \& B_2, b)]$$

$\wedge$

$$\forall B \in \text{Bool}. [Q(B, \text{true}) \implies Q(\neg B, \text{false})]$$

$\wedge$

$$\forall B \in \text{Bool}. [Q(B, \text{false}) \implies Q(\neg B, \text{true})]$$

$\wedge$

$$\forall b \in \mathbb{B}. [Q(b, b)]$$

$\wedge$

$$\forall B_1, B_2 \in \text{Bool}. \forall b \in \mathbb{B}. [Q(B_2 \& B_1, b) \implies Q(B_1 \& B_2, b)]$$

$\implies$

$$\forall B \in \text{Bool}. \forall b \in \mathbb{B}. [B \Downarrow_e b \implies Q(B, b)]$$